PRACTICE PROGRAMS

Q1 Matrix multiplication

10 marks

Matrix multiplication

Your Code

```
public class multiplication{
    public static void main(String[]args){
        int[][]a={
            {1,2},
            {3,4}
        };
        int[][]b={
            {5,6},
            {7,8}
        };
        int[][]c= new int[2][2];
        for(int i=0;i<2;i++){
            for(int j=0;j<2;j++){
                for(int k=0;k<2;k++){
                    c[i][j]+=a[i][k]*b[k][j];
                }
            }
        }
        System.out.println("result");
        for(int i=0;i<2;i++){
            for(int j=0;j<2;j++){
                System.out.print(c[i][j]+ " ");

            }
            System.out.println();
        }
    }
}
```

Input

stdin input

Output

```
result
19 22
43 50
```

Q2 . Largest number

10 marks

. Largest number

Your Code

```
public class largest{
    public static void main(String[]args){
        int[]arr={14,30,6,8,9};
        int largest=arr[0];
        for(int i=1;i<arr.length;i++){
            if(arr[i]>largest){
                largest=arr[i];
            }
        }
        System.out.println("largest: "+largest);
    }
}
```

Input

stdin input

Output

```
largest: 30
```

Q3 **Smallest number** 10 marks

Smallest number

Your Code

```
public class smallest{
    public static void main(String[]args){
        int[]arr={14,30,6,8,9};
        int smallest=arr[0];
        for(int i=1;i<arr.length;i++){
            if(arr[i]<smallest){
                smallest=arr[i];
            }
        }


    System.out.println("smallest: "+smallest);
    }
}
```

Input

```
stdin input
```

Output

```
smallest: 6
```

Q4 **. Nth largest number** 10 marks

. Nth largest number

Your Code

```
import java.util.Arrays;
public class main{
    public static void main(String[]args){
        int[]arr={14,30,6,8,9,12};
        int n=2;
        Arrays.sort(arr);
        System.out.println(n+" th largest="+arr[arr.length-n]);
    }
}
```

Input

```
stdin input
```

Output

```
2 th largest=14
```

Q5 **Palindrome number/string** 10 marks

Palindrome number/string

Your Code

```
public class main{
    public static void main(String[]args){
        String str="madam";

        String rev="";
        for(int i=str.length()-1;i>=0;i--){
            rev=rev+str.charAt(i);

        }
        if(str.equals(rev))
            System.out.println("palindrome");
        else
            System.out.println("not a palindrome");
    }
}
```

Input

```
stdin input
```

Output

```
palindrome
```

Q6 **Removing duplicates in array** 10 marks

Removing duplicates in array

Your Code

```
public class main{
    public static void main(String[]args){
        int[]arr={1,2,2,3,4,4,5};
        System.out.println("array after removing:");
        for(int i=0;i<arr.length;i++){
            boolean duplicate=false;
            for(int j=0;j<i;j++){
                if(arr[i]==arr[j]){
                    duplicate=true;
                    break;
                }
            }
            if(!duplicate){
                System.out.print(arr[i]+" ");
            }
        }
    }
}
```

Input

stdin input

Output

```
array after removing:
1 2 3 4 5
```